Concepts Clarification

C5b Python with SQLite

A. Can PK be NULL?

Create a table using only TEXT fields

```
CREATE TABLE OnlyText (
        Name TEXT PRIMARY KEY,
        Class TEXT);
```

Can we insert records by not providing the values for some fields?

Yes, if the field is not a Primary Key (PK) and there is no NOT NULL constraint.

A Null value will be indicated.

Insert one data with 2 fields:

```
INSERT INTO OnlyText
        (Name, Class)
        VALUES  ('Name1', 'Class1');
```

Insert multiple data with 2 fields:

```
INSERT INTO OnlyText
        (Name, Class)
        VALUES  ('Name2', 'Class2'),
                ('Name3', 'Class3'),
                ('Name4', 'Class4');
```

Omit the values for `Class` field:

```
INSERT INTO OnlyText
        (Name)
        VALUES  ('Name5'),
                ('Name6'),
                ('Name7'),
                ('Name8');
```

	Name	Class
	Filter	Filter
1	Name1	Class1
2	Name2	Class2
3	Name3	Class3
4	Name4	Class4
5	Name5	NULL
6	Name6	NULL
7	Name7	NULL
8	Name8	NULL

Can I omit the values for the PK field?

Omit the values for PK `Name` field:

```
INSERT INTO OnlyText
       (Class)
       VALUES  ('Class5'),
               ('Class6'),
               ('Class7'),
               ('Class8');
```

How come I can still insert records without providing the values for the PK field?

	Name	Class
	Filter	Filter
1	Name1	Class1
2	Name2	Class2
3	Name3	Class3
4	Name4	Class4
5	Name5	NULL
6	Name6	NULL
7	Name7	NULL
8	Name8	NULL
9	NULL	Class5
10	NULL	Class6
11	NULL	Class7
12	NULL	Class8

This is because SQLite does not enforce the NOT NULL constraint strictly for the PK field

But other DBMS will enforce this strictly.

B. Auto-Increment

C5a Python with SQLite

Slide 8 – Why we can insert record without specifying the value for the Id?

Execute SQL

Insert one row of data into table




```
INSERT INTO Products
      (ProductName, UnitPrice, Package, SupplierName, SupplierCountry)
VALUES
      ('Chai', 18, '10 boxes x 20 bags', 'Exotic Liquids', 'UK');
```

Id	ProductName	UnitPrice	Package	SupplierName	SupplierCountry
1	Chai	18	10 boxes x 20 bags	Exotic Liquids	UK

- `Id` field will be populated by auto-increment generated values

Table with INTEGER `Id` as the PK

```
CREATE TABLE NoAuto (
        Id INTEGER PRIMARY KEY,
        Name TEXT);
```

Insert multiple data with 2 fields:

```
INSERT INTO NoAuto
        (Id, Name)
        VALUES  (1, 'Name1'),
                (2, 'Name2'),
                (3, 'Name3'),
                (4, 'Name4');
```

Omit the values for PK `Id` field:

```
INSERT INTO NoAuto
        (Name)
        VALUES  ('Name5'),
                ('Name6');
```

SQLite will auto generate the values for the INTEGER `Id` PK field.

But other DBMS may not do so.

What if I omit the values for the INTEGER `Id` PK field?

	Id	Name
	Filter	Filter
1	1	Name1
2	2	Name2
3	3	Name3
4	4	Name4
5	5	Name5
6	6	Name6

Table using Auto-Increment for PK

```
CREATE TABLE WithAuto (
        Id INTEGER PRIMARY KEY AUTOINCREMENT,
        Name TEXT);
```

Insert multiple data with Auto-Increment for PK:

```
INSERT INTO WithAuto
        (Name)
        VALUES  ('Name1'),
                ('Name2');
```

Insert a data with the Id field:

```
INSERT INTO WithAuto
        (Id, Name)
        VALUES  (5, 'Name5');
```

What if I specify the Id value?

	Id	Name
	Filter	Filter
1	1	Name1
2	2	Name2
3	5	Name5

Auto-Increment Id values will still be generated subsequently.

Insert multiple data with Auto-Increment for PK:

```
INSERT INTO WithAuto
        (Name)
        VALUES  ('Name6'),
                ('Name7'),
                ('Name8');
```

	Id	Name
	Filter	Filter
1	1	Name1
2	2	Name2
3	5	Name5
4	6	Name6
5	7	Name7
6	8	Name8

Will the Id value be 'recycled'?

Delete the last record from `NoAuto` table:

```
DELETE FROM NoAuto WHERE Id=6;
```

Insert some records without the Id field:

```
INSERT INTO NoAuto
        (Name)
        VALUES  ('Name7'),
                ('Name8'),
                ('Name9');
```

What if I delete the last record, will that Id value be 'recycled' and used again?

	Id	Name
	Filter	Filter
1	1	Name1
2	2	Name2
3	3	Name3
4	4	Name4
5	5	Name5
6	6	Name7
7	7	Name8
8	8	Name9

Yes, but this is no good!!!
Id 6 is supposed to refer to the person being deleted.

Delete the last record from `WithAuto` table:

```
DELETE FROM WithAuto WHERE Id=8;
```

Insert some records without the Id field:

```
INSERT INTO WithAuto
        (Name)
        VALUES  ('Name9'),
                ('Name10');
```

No, and this is good!!!
Id 8 is not used.

	Id	Name
	Filter	Filter
1	1	Name1
2	2	Name2
3	5	Name5
4	6	Name6
5	7	Name7
6	9	Name9
7	10	Name10

C. Deletion Denied

C4-2 Basic SQL

Slides 36-38 – Why can't we delete the records or drop the table?

Tables with Foreign Key (FK)

Table Description:

```
Customer(Id, Name)
Purchase(Product, Cid*)
```

PK: ___
FK: *

```
CREATE TABLE Customer (
        Id INTEGER PRIMARY KEY AUTOINCREMENT,
        Name TEXT);

CREATE TABLE Purchase (
        Product TEXT,
        Cid INTEGER References Customer(Id));
```

Insert some data:

```
INSERT INTO Customer
        (Name)
        VALUES  ('Adam'),
                ('Bob');

INSERT INTO Purchase
        (Product, Cid)
        VALUES  ('Cake', 2);
```

Table: Customer

	Id	Name
	Filter	Filter
1	1	Adam
2	2	Bob

Table: Purchase

	Product	Cid
	Filter	Filter
1	Cake	2

Deletion Allowed or Denied?

Delete a customer:

```
DELETE FROM Customer WHERE Id=1;
DELETE FROM Customer WHERE Id=2;
```

Table: Customer

	Id	Name
	Filter	Filter
1	1	Adam
2	2	Bob

Table: Purchase

	Product	Cid
	Filter	Filter
1	Cake	2

Can I delete any customer?

Customer with Id=2 cannot be deleted.

FOREIGN KEY constraint failed

Referential Integrity has to be enforced.

Foreign Key (FK) helps to maintain the data integrity by preventing erroneous deletion.

Important to know

You must create the tables correctly based on the foreign keys specified in the table description.

```
Customer (Id, Name)
Purchase (Product, Cid*)
```

Referential Integrity: This means that for every record in a Child table, there must be a valid matching record referencing to the Parent table.

Due to this, the Parent table must be populated first, before populating the Child table.

D. Single or Double Quote?

C5b Python with SQLite

Slide 10 – When to use single quote ' and when to use double quote " ?

Execute SQL

Insert multiple rows of data into table




```
INSERT INTO Products
      (ProductName, UnitPrice, Package, SupplierName, SupplierCountry)
VALUES
      ('Chang', 19, '24 - 12 oz bottles', 'Exotic Liquids', 'UK'),
      ('Aniseed Syrup', 10, '12 - 550ml bottles', 'Exotic Liquids', 'UK'),
      ("Chef Anton's Cajun Seasoning", 22, '48 - 6 oz jars', 'New Orleans
Cajun Delights', 'USA'),
      ("Chef Anton's Gumbo Mix", 21.35, '36 boxes', 'New Orleans Cajun
Delights', 'USA');
```

Use double quote " if data contains single quote '

When to use single or double quote?

If a string contains single quotation mark within it, then use double quotes to enclose the string. (Conversely, use single quotes to enclose the string that has double quotation marks within it.)

For example:

```
String1 = "This is Mr Lim's bag."
String2 = 'He says: "Hello World."'
```

If a string contains both the single and double quotation marks within it, then use either triple-single or triple-double quotes to enclose the string.

For example:

```
String3 = ''' Mr Lim's name is "John".'''
String4 = """ Mr Lim's name is "John"."""
```

E. Parameterise an SQL command

C5a Python with SQLite

Slides 11-16 – Using **?** as the parameter(s) for the data value(s) when inserting or updating data into the database.

Updating the value of a field

Refer to C5a Python with SQLite Slide 14 to *change Thomas Hardy's Country from UK to Singapore*, instead of writing the command as:

```
c.execute('''UPDATE Customers SET Country=?\
WHERE FirstName="Thomas" \
AND LastName="Hardy";''', ('Singapore',))
```

We could have written it as:

```
c.execute('''UPDATE Customers SET Country='Singapore'\
WHERE FirstName="Thomas" \
AND LastName="Hardy";''')
```

But in this way of hardcoding, we will not be able to get the user to input their choice of country.

Use user's input for updating the value of a field

The following code allows the user to input the `NewCountry` for Thomas Hardy:

```
NewCountry = input("Enter a new country: ")

c.execute('''UPDATE Customers SET Country=?\
WHERE FirstName="Thomas" \
AND LastName="Hardy";''', (NewCountry,))
```

Note that the structure of the `execute()` command line is:

```
c.execute(sql_string, data_tuple)
```

We use the question mark (?) as the parameter(s) for the data value(s) to be **inserted** or **updated** in the database.

The End

C5b Python with SQLite – Concepts Clarification